Namma Ooru AI தமிழ்நாடு

Smart Community Governance Platform • Software Development Life Cycle (SDLC)

Simple SDLC Report

Version 1.0 • October 2026

1. **Planning** — Civic problem & goal
2. **Requirements** — Voice, Tamil & AI
3. **Design** — Clean UI/UX & Icons
4. **Coding** — React + FastAPI
5. **Testing** — Voice, Tamil & Build
6. **Deploy** — Live on Localhost

Step 1 Planning (Idea & Purpose)

- **Citizen Problem:** Citizens struggle with uncollected garbage, damaged roads, water supply leaks, broken streetlights, and clogged drainage.
- **Platform Goal:** Build *Namma Ooru AI* — an accessible smart governance portal for all 25 Corporations and urban bodies across Tamil Nadu.
- **Core Value:** Enable 60-second problem reporting through voice or form in Tamil and English with transparent resolution tracking.

Step 2 Requirement Analysis

- **Speech-to-Text:** Web Speech API capturing voice in Tamil (`ta-IN`) and English (`en-IN`) for senior citizens.
- **AI Auto-Triaging:** Classify grievances and route immediately to Sanitation, PWD, Water Board, or Electrical departments.
- **OpenStreetMap:** Pinpoint exact GPS coordinates and reverse-geocode ward location.
- **Live Tracking ID:** Unique tracking numbers (e.g., `NOAI-2026-1001`) with real-time milestone timeline.
- **Role Dashboards:** Citizen tracking & Municipal Officer dispatch centers.

Step 3 Design (UI / UX Architecture)

- **Theme & Aesthetics:** Clean white civic background with rich green and blue accents (WCAG 2.1 AA accessible).
- **Home Page Services Hub:** Big 1-click icon cards on the homepage for Report, Track, Dashboard, Officer, Login, About, and 1913 Helpline.
- **Clutter-Free Navigation:** Clean, focused sticky header featuring Brand identity, 1-click Home return, and language switcher.
- **Responsive Layout:** Flawless mobile, tablet, and desktop viewing.

Step 4 Implementation (Tech Stack)

- **Frontend:** React 18 + Vite (fast component architecture, 1.98s production build time).
- **Backend Engine:** Python 3.12 + FastAPI (asynchronous REST APIs, Pydantic v2 validation).
- **Maps:** OpenStreetMap + Leaflet (interactive pin drop without commercial API paywalls).
- **Bilingual i18n:** Complete dictionary switching between English and Tamil (தமிழ்).
- **Icons:** Lucide React modern SVG icon library.

Step 5 Testing & Verification

- **Voice Input Test:** Microphone accurately captures voice and populates problem descriptions in Tamil and English.
- **Language Switch Test:** Verified 100% of labels, headers, and buttons switch correctly into Tamil script.
- **Metric Reconciliation:** Verified live counts (Reported: 2, Pending: 2, In Progress: 2, Resolved: 1) match database records.
- **Production Build Test:** `npm run build` passed cleanly with zero errors.

Step 6 Deployment (Running Live)

- **Frontend Server:** React running on `http://localhost:5173`
- **Backend API Server:** FastAPI running on `http://localhost:8000`
- **Interactive API Docs:** Swagger UI available at `http://localhost:8000/docs`
- **Ready for Cloud:** Container-ready blueprint for Google Cloud Run and Cloud SQL (PostgreSQL).

Frontend:	API Server:	Swagger Docs:	Status: Live & Functional
http://localhost:5173	**http://localhost:8000**	**http://localhost:8000/docs**	